Introduction

This report is about the Springfield-based doughnut company, Lard & Land. The company believes that data may have been taken – specifically the recipe for their "Honey Duff Donuts". The lead suspect is someone under the alias "Ms Taurus Smith". The evidence supplied were a liquid damaged mobile phone, a USB flash drive, a forensic image of a laptop, a NTUSER.DAT file and a network PCAP file. The general aim will be to identify if this "Ms Taurus Smith" has accessed and distributed confidential information, interacted over networks, see what their next actions may be and see if there are any other parties involved.

Evidence Identification

Exhibit ID	Description	Source	Date Received
A	Liquid-damaged mobile phone	742 Evergreen Terrace	15/12/2025
B	USB Flash Drive	742 Evergreen Terrace	15/12/2025
C	Laptop Forensic Image (Laptop_Image.001)	USB drive	15/12/2025
D	NTUSER.DAT	Shop computer	16/12/2025
E	Network PCAP files	Company network	12/12/2025

Chain of custody

Date	Person	Action	Location
15/12/2025	Police officer A	Seized laptop	742 Evergreen Terrace
17/12/2025	Forensic Analyst Caiden Harries	Created forensic image	Digital Forensics Lab
17/12/2025	Forensic Analyst Caiden Harries	Verified hash	Digital Forensics Lab

The forensic image was loaded into Autopsy as a disk image. Autopsy automatically detected and analysed the available disk partitions contained within the image. Where applicable, both allocated and unallocated space within the partition(s) were examined as part of the analysis.

The forensic disk image Laptop_Image.001 was verified prior to analysis using MD5, SHA-1 and SHA-256 hashing. These values were recorded and used to confirm the integrity of the evidence during the investigation. All examination was carried out using forensically-sound, read-only tools to ensure that the original evidence was not altered.

MD5: cf2f57f21f7604a5c47ca94bce2c3b66

SHA1: 4aa91294fecc1ad92c5b4f34ac5e02551efea6b7

SHA256: 4588f113b6ad1dcfe1c631f7dcc0ebe0c7ed94dc6216ed7d0ca3b914e4c5e4a4

The NTUSER.DAT registry hive was hashed prior to analysis using MD5 and SHA-256 hashing. These values were recorded and used to confirm the integrity of the evidence during the investigation. All examination was carried out using forensically-sound, read-only tools to ensure that the original evidence was not altered.

MD5: 288a9cd8f18de7d8626ccdbc39eb1e6c

SHA-256: b65afcc7caf389a10d82bb5206f448da25b07b179e31c56aae73ff735bfc0b80

Tools

- Autopsy - disk image and analysis, NTUSER.DAT analysis
- Wireshark - analysed PCAP network traffic

Methodology

The primary user account analysed throughout this methodology was “Taurus Smith”.

- Cryptographic hash values were used to verify the integrity of the image prior to and throughout the investigation.
- All analysis was performed on the forensic image, not the original device, using read-only access to avoid any modification of evidence.
- Created read only forensic image, verified hashes
- Loaded image into autopsy, ran ingest modules
- Examined file system, deleted files, USB history, browser history, emails
- Analysed NTUSER.DAT for user activity
- Analysed PCAP files in Wireshark (filters etc.)
- Built a timeline correlating these events

Forensic Analysis

Laptop Image Analysis

Deleted File Recovery Findings:

Examining the deleted files (figure 1) under the directory for “Taurus Smith” (the employee alias that Lard & Land believes may have committed malfeasance) revealed several highly relevant items. Firstly, we must mention that all of these files mentioned were tagged with Flags (dir) unallocated and Flags (Meta) unallocated. Files recovered from unallocated space indicate they were previously deleted. This suggests an attempt to remove traces of activity. To start with, perhaps the most important part, we can see the files “Lardland Super Donut Ingredients.png”, “Lardland Super Donut Instructions.pdf” and multiple versions of “ring recipe.txt”. All three of these directly relate to the donut recipes belonging to Lard & Land and their possibly their “Honey Duff Donuts

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
Lardland Super Donut Ingredients.png				2010-03-31 18:55:06 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 18:56:43 BST	940161	Unallocated	Unallocated
~$Digital Evidence Standards (Public).ppt				2010-03-07 20:22:08 GMT	0000-00-00 00:00:00	2010-03-07 00:00:00 GMT	2010-03-07 20:21:50 GMT	165	Unallocated	Unallocated
New folder				2010-03-31 18:57:46 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 18:57:45 BST	0	Unallocated	Unallocated
Penelope Olsen				2005-01-01 18:50:02 GMT	0000-00-00 00:00:00	2010-03-08 00:00:00 GMT	2004-03-08 00:17:47 GMT	0	Unallocated	Unallocated
Lardland Super Donuts Instructions.pdf				2010-03-31 18:55:06 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 18:58:38 BST	4489177	Unallocated	Unallocated
ip-crime-report-2019-20.pdf				2010-03-31 18:55:06 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 18:58:44 BST	14930788	Unallocated	Unallocated
ring recipe.txt.sb-ace39f49-PiYykM				2022-11-06 10:58:50 GMT	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:55:14 BST	3522	Unallocated	Unallocated
ring recipe.txt - Copy.sb-ace39f49-PiYykM				2010-03-31 19:56:22 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:55:22 BST	12	Unallocated	Unallocated
sb-ace39f49-PiYykM				2022-11-06 10:58:50 GMT	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:55:14 BST	3522	Unallocated	Unallocated
_INGDN~1.RTF				2009-11-06 19:12:52 GMT	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:21:22 BST	2694	Unallocated	Unallocated
pdf.txt.txt				2010-03-07 21:24:36 GMT	0000-00-00 00:00:00	2010-03-07 00:00:00 GMT	2010-03-07 21:24:11 GMT	4	Unallocated	Unallocated
_ARDIF~1.JPE				2022-11-06 19:09:52 GMT	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:25:18 BST	352587	Unallocated	Unallocated
New Text Document.txt				2010-03-07 21:24:12 GMT	0000-00-00 00:00:00	2010-03-07 00:00:00 GMT	2010-03-07 21:24:11 GMT	0	Unallocated	Unallocated
Steg-1.1.0.0-Windows-x64.zip				2009-11-30 10:33:26 GMT	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 19:01:28 BST	30291561	Unallocated	Unallocated
_ravel.png				2010-03-31 19:12:52 BST	0000-00-00 00:00:00	2010-03-31 00:00:00 BST	2010-03-31 20:06:39 BST	636468	Unallocated	Unallocated
New folder				2022-03-08 00:10:04 GMT	0000-00-00 00:00:00	2022-03-08 00:00:00 GMT	2022-03-08 00:10:02 GMT	0	Unallocated	Unallocated
important email.eml				2022-03-08 06:27:06 GMT	0000-00-00 00:00:00	2022-03-08 00:00:00 GMT	2022-03-08 00:13:38 GMT	2881	Unallocated	Unallocated

Figure 1 – deleted files

Along with this, we are able to see multiple files with the suffix “.sb-ace39f49-PiYykM”. These filenames look like hidden backup files or temporary copies. This suggests the suspect tried to hide or make copies of the recipe documents before trying to delete them. Furthermore, we can see a tool installed from the file “Steg-1.1.0.0-Windows-x64.zip” was found among the deleted files. The presence of steganography software indicates the capability to conceal data and may suggest attempts to hide sensitive material.

A deleted folder referencing one “Penelope Olsen” was recovered. This suggests the involvement of an additional individual being involved since this was a file deleted by “Taurus Smith”. However, there are no files deleted by this “Penelope Olsen” – we shall look as to why this is interesting later.

File System Examination

A review of the user content under “Documents and Settings/Taurus Smith” identified multiple donut-related image files which appear to relate to donut products. These included files such as “Lardland Super Donut Ingredients.png”, “good donut.png”, “ingredients.png” and “baked_donut_recipe_featured.jpg.”.

Many of these files (figure 2) were located under Taurus Smith were under the directories of “Desktop”, “Recent” and “SendTo”. This indicates that the suspect has opened, viewed and possibly / prepared to distribute these files for transfer. This demonstrates active handling of proprietary recipe information belonging to Lard & Land. The coexistence of both allocated (active) and unallocated (deleted) copies of recipe-related files suggests repeated access followed by attempted deletion and concealment.



Figure 2 – images stored by Taurus Smith

Email Communications Analysis

```
X-Account-Key: account3
Return-Path: <useram@domaingal.com>
Received: from cluster.serviceorg.com (cluster.serviceorg.com [10.213.4.42])
        by mail.serviceorg.com (8.12.11.20100614/8.12.11) with ESMTP id kARABqg0023308
        for <userpf@domaingal.com>; Thu, 04 Mar 2010 11:11:54 +0100
Received: from cluster.serviceorg.com (localhost.localdomain [127.0.0.1])
        by localhost.serviceorg.com (Postfix) with ESMTP id 30C6248047
        for <userpf@domaingal.com>; Thu, 04 Mar 2010 11:11:52 +0100 (CET)
Received: from spiderwall.localdomain
        (gateway-gal.domaingal.com [10.10.254.4])by
        cluster.serviceorg.com (Postfix) with SMTP id 5530848056for
        <userpf@domaingal.com>; Thu, 04 Mar 2010 11:11:48 +0100
        (CET)
Received: (qmail 27268 invoked from network); 27 Nov 2010 10:11:48 -0000
Received: from pc-seg.dominio.locale (HELO PCSEG)
        (10.10.0.100) by 12.45.87.123 with SMTP; 27 Nov 2010 10:11:47 -0000
Message-ID: <002601c7120c$7220d3d0$6400670a@dominio.locale>
From: "name surname" <useram@domaingal.com>
To: <destuser@domaingal.com>
Subject: determ- iter
Date: Thu, 04 Mar 2010 11:11:46 +0100
MIME-Version: 1.0
Content-Type: multipart/alternative;
        boundary="----=_NextPart_000_0023_01C71214.D3D49A00"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2900.2869
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2900.2962
X-imss-version: 2.044
X-imss-result: Passed
X-imss-scanInfo: M:P L:E SM:0
X-imss-tmaseResult: TT:0 TS:0.0000 TC:00 TRN:0 TV:3.6.1039(14838.003)
X-imss-scores: Clean:64.23839 C:2 M:3 S:5 R:5
X-imss-settings: Baseline:1 C:4 M:4 S:4 R:4 (0.0000 0.0000)
X-Scanned-By: MIMEDefang 2.54 on 10.213.4.47
VGhpcyBpcyBhIG11bHRpLXBhcnQgbWVzc2FnZSBpbiBNSU1FIGZvcm1hdC4KCi0tLS0tLT1fTmV4dFBhcnRfMDA
ZIci1FbmNvZGluZzogcXVvdGVkLXByaW50YWJsZQoKUGxlYXNlIGdvIHRvIHRoZSB3ZWJzaXRlIGFuZCBkZWNvZC
CgotLS0tLS09X05leHRQYXJ0XzAwMF8wMDIzXzAxQzcxMjE0LkczRDQ5QTAwCkNvbnRlbnQtVHlwZTogdGV4dC
UFVCTEIDICItLy9XM0MvL0RURCBIVE1MIDQuMCBUcmFuc2l0aW9uYWwvL0VOIj4KPEhUTUw+PC9qvV
PTNEIk1TSFRNTC A2LjAwLjI5MDAuNjk2MyIgbmFtZT0zREdFTkVSQVRPUj4KPFNUWUxFPjwvU1RZTEU+CjwvSE
2IsbCBzZWUgdGhlIGRldGFpbHMgMGlGh0dHBzOi8vc3R5bGVzdXh4LmdpdGh1Yi5pby9tdGVnYW5vZ3JhcGh5Lwo8
Expected Results:
you need to decode the contents.
```

Figure 3 – suspicious email

A suspicious email message (figure 3), marked “important email.eml” and given the subject “determ- iter” was recovered from the subject’s system. The email originated from “customer@domaingal.com” and was addressed to “detuser@domaingal.com”. The message body contained a large block of encoded text followed by instructions: “Final Result: You need to decode the contents.”

The presence of encoded data and an explicit instruction to decode the file is most likely indication of deliberately hiding communications. This suggests the email was used as a method to conceal data and transfer it. It is my understanding that this may likely be the stolen recipe information. This conclusion is further supported by the discovery of steganography software and deleted recipe documents.

Browser History

Analysis of the browser history reveals significant browsing activity. Whilst a large quantity was under another user, there were several that were not marked by any username. Of these, we can see payment-based websites such as eBay, PayPal and Access.com. Many entries correspond to image, CSS and script resources, indicating normal website rendering. Google search history reflected queries relating to digital communication methods (ICQ, IRC), data compression tools (WinZip), and computer forensics, as well as less relevant entertainment-related searches. These findings indicate that the suspect was technically capable and familiar with online tools for communication and data handling.

Recent Document Analysis

Examination of the Decent Documents revealed multiple .lnk shortcut references to image and document files within the shared folders, including “\Kwarren-laptop\Documents\The Precious\”. This link indicates the files were opened by an authenticated user prior to image acquisition. Other entries showed Windows Media Player activity, confirming a user has viewed many files. Furthermore, several NTUSER.DAT entries were listed within the Recent Documents list, consistent with active user registry interaction.

USB Device Activity

The USB device attached section (figure 4) revealed the multiple recent USB mass-storage devices connected to the system over the documented period. Device time-stamps overlap with the suspected timeframes of data theft. This presence of multiple storage devices may indicate the transfer of confidential recipe files from the workstation to external media.

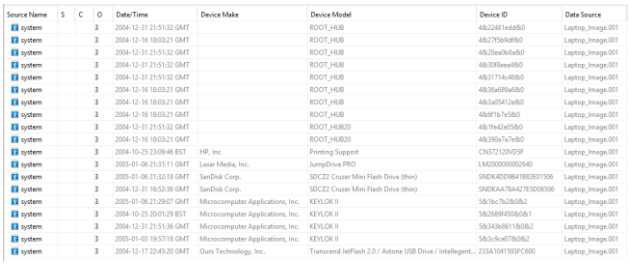


Figure 4 – recent devices attached to the suspect’s laptop

NTUSER.DAT Registry Analysis

Typed URLs

The NTUSER.DAT file was extracted and analysed in Autopsy as a standalone logical file. This file contains the per-user Windows Registry hive and records user-specific activity including Internet Explorer typed URL history. Within the registry path "HKEY_CURRENT_USER\Software\Microsoft\Internet Explorer\TypedURLs" the following URLs were found: "http://131.10.28.251:53/" (figure 5) and "http://cardiffdonut.com/fwlink/?LinkId=69157".

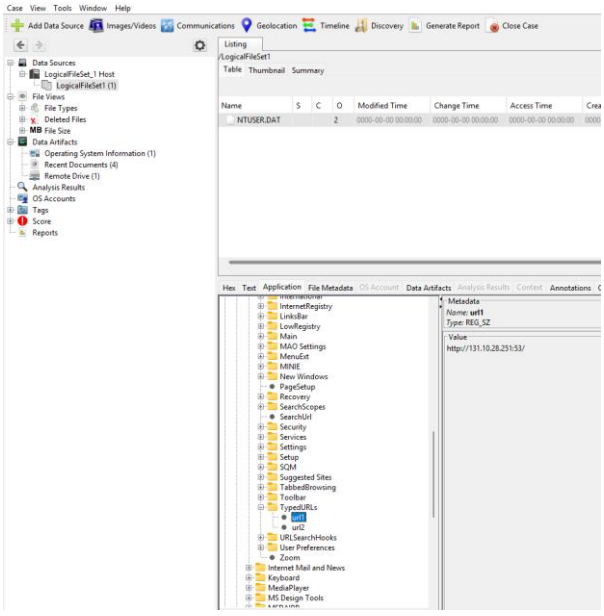


Figure 5 – the typed URL 1

These entries indicate websites that were manually typed into the Internet Explorer address bar by the user. Typed URLs are user-initiated entries, meaning the suspect manually navigated to these web locations. Typed URLs are strong evidence for intent rather than automatically generated website links. Both URLs appear to relate to the investigation context - the IP address suggests direct access to a server, while the cardiffdonut.com link reflects navigation to an external website associated with the case. This evidence supports the conclusion that the user knowingly navigated to these locations.

Recent Documents

The NTUSER.DAT registry hive was examined in Autopsy, separately from the laptop image. Registry artefacts relating to user activity were extracted using the Recent Activity and Registry ingest modules. From this we were able to see a few things. For one, these were found under “Recent Documents”:

- Undercover-Agents-List-For-United-Kingdom.xlsx
- Undercover-Agents-List-For-United-States.xlsx
- CC R&D Backstopped Accounts.xlsx
- Credit-Card-Numbers-For-Research.xlsx

Autopsy reports these entries as “Recently opened according to Office MRU”. This indicates the user actively opened and viewed these documents. These files contain highly sensitive agent and financial data, providing strong evidence that the suspect accessed classified material.

Excel Access

Program execution history was recovered from the UserAssist key “HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist". The ROT-13 decoded entry …\Office14\EXCEL.EXE showed a RunCount value equivalent to 116 executions, indicating Microsoft Excel had been opened 116 times by the user. This indicates that the user has access business-sensetive content several times consistently, perhaps handling recipes, finances or employee data, and we can assume by the count it is not a one-off accident.

Wireshark PCAP Analysis

Exhibit B

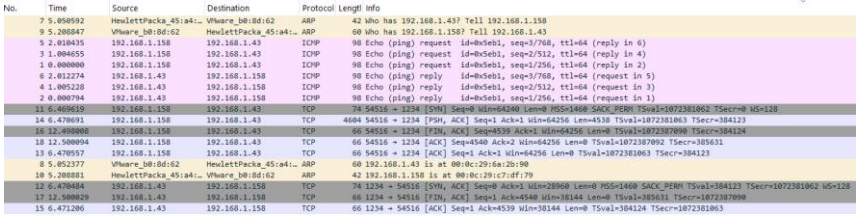
Using this Wireshark capture, we are able to see that the suspect’s laptop (192.168.1.158) communicates with an external server via HTTPS at 64.12.24.50. The traffic is labelled as encrypted. However, the packets also contain clear-text human conversation. The sender admits to downloading “secret project file” and copying to a USB drive to be handed over. This message proves the intention of the suspect and confirming that confidential data was taken with intent and prepared for transfer.

Exhibit C

Exhibit C shows us another Wireshark capture of communications between the suspect laptop (192.168.1.158) and an external server (64.12.24.50). Although labelled as secure HTTPS traffic, the packet data contains readable conversation text. The suspect confirms travel plans to Hawaii and states they will bring a USB drive. This shows that the suspect not only copied confidential data to removable storage but also planned to transfer it to another person during a meeting. This demonstrates intent, coordination, and prior planning.

Exhibit D

Network traffic was analysed using Wireshark (figure 6). The suspect laptop was identified as IP 192.168.1.158, based on repeated outbound ICMP and TCP traffic originating from this address. The suspect laptop communicated directly



No.	Time	Source	Destination	Protocol	Length	Info
7	5.050592	HewlettPacka_45:a4:…	VMware_b0:8d:62	ARP	42	Who has 192.168.1.43? Tell 192.168.1.158
9	5.208847	VMware_b0:8d:62	HewlettPacka_45:a4:…	ARP	60	Who has 192.168.1.158? Tell 192.168.1.43
5	2.010435	192.168.1.158	192.168.1.43	ICMP	98	Echo (ping) request id=0x5eb1, seq=3/768, ttl=64 (reply in 6)
3	1.004655	192.168.1.158	192.168.1.43	ICMP	98	Echo (ping) request id=0x5eb1, seq=2/512, ttl=64 (reply in 4)
1	0.000000	192.168.1.158	192.168.1.43	ICMP	98	Echo (ping) request id=0x5eb1, seq=1/256, ttl=64 (reply in 2)
6	2.012274	192.168.1.43	192.168.1.158	ICMP	98	Echo (ping) reply id=0x5eb1, seq=3/768, ttl=64 (request in 5)
4	1.005228	192.168.1.43	192.168.1.158	ICMP	98	Echo (ping) reply id=0x5eb1, seq=2/512, ttl=64 (request in 3)
2	0.000794	192.168.1.43	192.168.1.158	ICMP	98	Echo (ping) reply id=0x5eb1, seq=1/256, ttl=64 (request in 1)
11	6.469619	192.168.1.158	192.168.1.43	TCP	74	54516 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1072381062 TSecr=0 WS=128
14	6.470691	192.168.1.158	192.168.1.43	TCP	4604	54516 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=4538 TSval=1072381063 TSecr=384123
16	12.498008	192.168.1.158	192.168.1.43	TCP	66	54516 → 1234 [FIN, ACK] Seq=4539 Ack=1 Win=64256 Len=0 TSval=1072387090 TSecr=384124
18	12.500094	192.168.1.158	192.168.1.43	TCP	66	54516 → 1234 [ACK] Seq=4540 Ack=2 Win=64256 Len=0 TSval=1072387092 TSecr=385631
13	6.470557	192.168.1.158	192.168.1.43	TCP	66	54516 → 1234 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1072381063 TSecr=384123
8	5.052377	VMware_b0:8d:62	HewlettPacka_45:a4:…	ARP	60	192.168.1.43 is at 00:0c:29:6a:2b:90
10	5.208881	HewlettPacka_45:a4:…	VMware_b0:8d:62	ARP	42	192.168.1.158 is at 00:0c:29:c7:df:79
12	6.470484	192.168.1.43	192.168.1.158	TCP	74	1234 → 54516 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=384123 TSecr=1072381062 WS=128
17	12.500029	192.168.1.43	192.168.1.158	TCP	66	1234 → 54516 [FIN, ACK] Seq=1 Ack=4540 Win=38144 Len=0 TSval=385631 TSecr=1072387090
15	6.471206	192.168.1.43	192.168.1.158	TCP	66	1234 → 54516 [ACK] Seq=1 Ack=4539 Win=38144 Len=0 TSval=384124 TSecr=1072381063

Figure 6 – network traffic exhibit D

with IP 192.168.1.43. Several ICMP request and reply packets confirm that both devices were online and reachable to one another.

The suspect laptop communicated directly with IP 192.168.1.43 and application-level data was exchanged between the two systems. This tells us that there was a direct connection between the suspect laptop and another device on the same network, supporting the idea that at least one other party was involved in the incident.

Exhibit E

Network capture Exhibit E was also analysed using Wireshark. This capture again confirmed active two-way traffic between 192.168.1.158 (suspect laptop) and 192.168.1.43 (second device). There were hundreds of echo-requests and echo-reply packets observable, confirming that both systems online and communicating with one another.

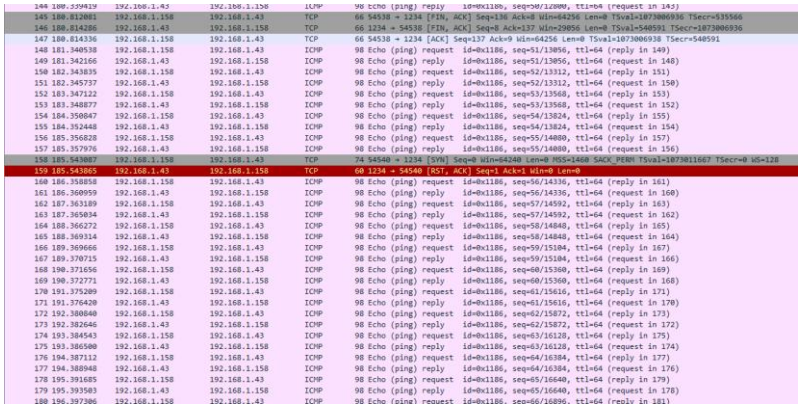
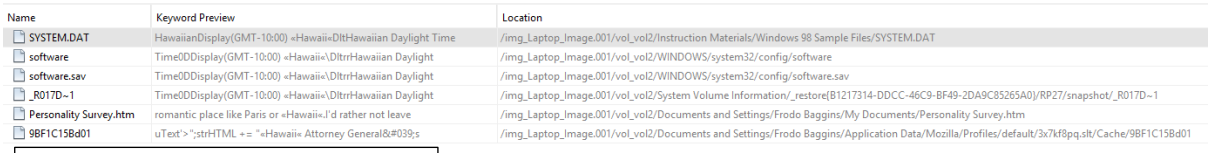


Figure 7 – network traffic exhibit E

Furthermore, a full TCP session was identified between the two systems. This demonstrates that an application-level session was successfully established and closed, meaning data was intentionally exchanged rather than accidental background traffic. The presence of a full TCP session means that the laptop was deliberately communicating with the other devices, rather than background communication. This further is proof of a 3rd party involved.

Planned Travel Destination

A keyword search for the term “Hawaii” was conducted. The term appeared in multiple system files including Windows registry hives referencing the Hawaiian time zone setting. A user-created file (“Personality Survey.htm”) also referenced Hawaii directly. Furthermore, we found a file referencing the “Hawaii Attorney General”. Combined with Exhibit C, which contains a message stating “see you in Hawaii”, these findings strongly indicate that Hawaii was the intended meeting or travel destination.



Name	Keyword Preview	Location
SYSTEM.DAT	HawaiianDisplay(GMT-10:00) «Hawaii«DltHawaiian Daylight Time	/img_Laptop_Image.001/vol_vol2/Instruction Materials/Windows 98 Sample Files/SYSTEM.DAT
software	Time0DDisplay(GMT-10:00) «Hawaii«\DltrrHawaiian Daylight	/img_Laptop_Image.001/vol_vol2/WINDOWS/system32/config/software
software.sav	Time0DDisplay(GMT-10:00) «Hawaii«\DltrrHawaiian Daylight	/img_Laptop_Image.001/vol_vol2/WINDOWS/system32/config/software.sav
_R017D~1	Time0DDisplay(GMT-10:00) «Hawaii«\DltrrHawaiian Daylight	/img_Laptop_Image.001/vol_vol2/System Volume Information/_restore{B1217314-DDCC-46C9-BF49-2DA9C85265A0}/RP27/snapshot/_R017D~1
Personality Survey.htm	romantic place like Paris or «Hawaii«.I'd rather not leave	/img_Laptop_Image.001/vol_vol2/Documents and Settings/Frodo Baggins/My Documents/Personality Survey.htm
9BF1C15Bd01	uText'>";strHTML += "«Hawaii« Attorney General's	/img_Laptop_Image.001/vol_vol2/Documents and Settings/Frodo Baggins/Application Data/Mozilla/Profiles/default/3x7kf8pq.slt/Cache/9BF1C15Bd01

Figure 8 – search term “Hawaii”

Other Individuals

While analysing the laptop image, we were able to see a deleted file named “Penelope Olsen” of which we are unsure whom this is. It is likely there is at least one more person involved due to the communications (discovered by the Wireshark analysis) from the suspect IP to the web servers and a second device (192.168.1.43), as well as direct TCP sessions and emails from the image. From this, we can conclude that this indicates deliberate collaboration with another person(s) who we can judge based off of the emails referencing Hawaii is the person the data may be given to.

Timeline Correlation

A correlation of evidence from the file system, registry, USB history and network captures was carried out to reconstruct the likely sequence of events:

- Ms Taurus Smith regularly accessed Excel sheets and other business-sensitive files. Evidence from NTUSER.DAT shows us Excel has been opened 116 times and recent document entries show us highly confidential information has been accessed by the user.
- Recipes and product files were then accessed and copied, such as “Lardland Super Donut Ingredients.png”, “Lardland Super Donut Instructions.pdf” and “ring recipe.txt” existed under the profile Ms Taurus Smith and had copies deleted, indicating repeated access followed by deletion
- Concealment applications were downloaded in order to attempt to cover up files, such as “Steg-1.1.0.0-Windows-x64.zip” and several hidden/backup file versions (including filenames ending .sb-ace39f49-PiYykM). These were then later deleted.
- External USB storage devices were connected
- The suspect communicated with a third party, seen via Wireshark. There is recorded HTTPS traffic containing text referencing downloading a “secret project file” and copying it to a USB drive, followed by an arranged meeting in Hawaii
- Direct TCP connection is established between IP addresses 192.168.1.158 (suspect laptop) and 192.168.1.43. A full TCP session was established and closed normally, confirming deliberate data exchange rather than background traffic.
- Several searches in registries, files and searches indicate references to Hawaii.

In summary, the combined evidence indicates that confidential recipes and other business-related matters were accessed and handled by “Ms Taurus Smith”. Removable media was available for transfer, whilst encrypted and hidden direct communication with a third party took place, discussing copies. Furthermore, the suspect arranged a meetup in Hawaii. These events together paint us a picture of someone who has prepared, accessed and distributed / prepared to distribute data whilst concealing communication.

Limitations

Whilst the forensic analysis was successfully conducted, have some limitations while analysing this data. Firstly, not all of the deleted data was able to be recoverable as data located in unallocated space is not the full content and not all of the context of all deleted files can be reconstructed.

Secondly, the PCAP was only a capture of a small period of communications and may not give the full picture limiting our information. Furthermore, the encrypted packets provide some data however there is still much unreadable, proving difficult in fully understanding what was being communicated due the entire time. The same goes with some HTTPS traffic.

Next, the USB devices do not say what data was copied across to it. Furthermore, we do not know which excel files Taurus Smith may have accessed. These limitations may not diminish the overall conclusion. However, recognition is necessary to maintain consistency and transparency.

Conclusion

We have several things to consider whilst concluding this analysis. Firstly, we are able to see very clearly via several means that the “Honey Duff Donuts” has been accessed by Ms Taurus Smith. We found several copies of this file deleted on their laptop image, and we can see the fact it has been accessed by them via said image. These were found in unallocated places and files recovered from unallocated space indicate they were previously deleted. This suggests an attempt to remove traces of activity. Furthermore, similar images were able to be seen via their personal computer. Alongside this, they have accessed Excel over 100 times, showing us that they have seen business-sensitive content consistently. This tells us very clearly the data has been accessed by Ms Taurus Smith.

Next, we are able to analyse the communications by Ms Taurus Smith. This was done via the laptop image via Autopsy, NTUSER.DAT and PCAP files via Wireshark. From this, we were able to pick up on many things. Firstly, we are aware of a third-party communication. The laptop (192.168.1.158) reached (with encryption) an external server via HTTPS at 64.12.24.50 and another direct IP via TCP traffic. Direct communication with 192.168.1.43 via TCP suggests that this was an intentional communication by the user and not just background noise, further proven by the sheer quantity of packets between the two.

Following this, we can see the laptop internet explorer history. Firstly, we can see it specifically typed the URL of a specific gateway into the address bar ("http://131.10.28.251:53/") alongside "http://cardiffdonut.com/fwlink/?LinkId=69157". This may indicate that these were typed manually rather than done by autocorrect and indicates accessing an external server as well as a possible rival donut company. Other internet results include payment services such as PayPal and Access.com hinting at payment for the release of the donut files, and private communication and file compression tools (ICQ, IRC and WinZip) alongside other computer forensic search terms. This may show us that the user is technically capable with online tools to do what they are suspected of doing.

Finally, whilst analysing these packets, we can see that there was a planned proposal to meet this third party in Hawaii due to the message "see you in Hawaii" which whilst being mostly encrypted had this left in plain-text. Another point to note would be the deletion of a folder named "Penelope Olsen" which may relate to another involved individual, although the exact role of this person cannot be confirmed from this artefact alone.

In conclusion, I believe that the person / alias “Ms Taurus Smith” has intentionally accessed this data, downloaded it and saved it to external USB devices. Next, they attempted to delete at least some of these files. Furthermore, intentional communication with outside via email and google searches show us that they intend to meet the third party in Hawaii. Based on the recovered evidence, there is very strong evidence that confidential company data was deliberately repeatedly accessed by the user and prepared for unauthorised access and sharing by the user “Taurus Smith” and that there is at least one other party involved in this.